# AI Impact on the Job Market (2024–2030)

## Executive Summary

Artificial Intelligence is expected to reshape labour markets, but its effects are unlikely to be uniform across industries. This analysis investigates whether AI exposure, automation risk, and industry characteristics are associated with projected employment growth through 2030.

## Research Questions:

Does automation risk predict job growth?

- Do AI-impacted occupations grow faster?
- Which industries benefit most from AI?
- Which industries are most vulnerable?

## Dataset & Methodology

This analysis examined approximately 30,000 occupations to evaluate whether AI exposure, automation risk, and industry characteristics influence projected employment growth between 2024 and 2030. Dataset (**AI Impact on Job Market: (2024–2030))** was gotten from Kaggle.

**Dataset Overview:** 30,000 occupations.

**Methods:** EDA, Feature Engineering, Correlation Analysis, ANOVA Testing, Industry Segmentation, and Growth Rate Analysis.

**Analytical Methods:** Exploratory Data Analysis, Correlation Analysis, ANOVA, Industry Segmentation, Growth Rate Analysis.

**Variables Analyzed:** AI Impact Level, Automation Risk, Salary, Experience, Remote Work Ratio, Job Openings (2024), Projected Openings (2030)

## Result

**Finding 1: Does Automation Risk Predict Growth?**

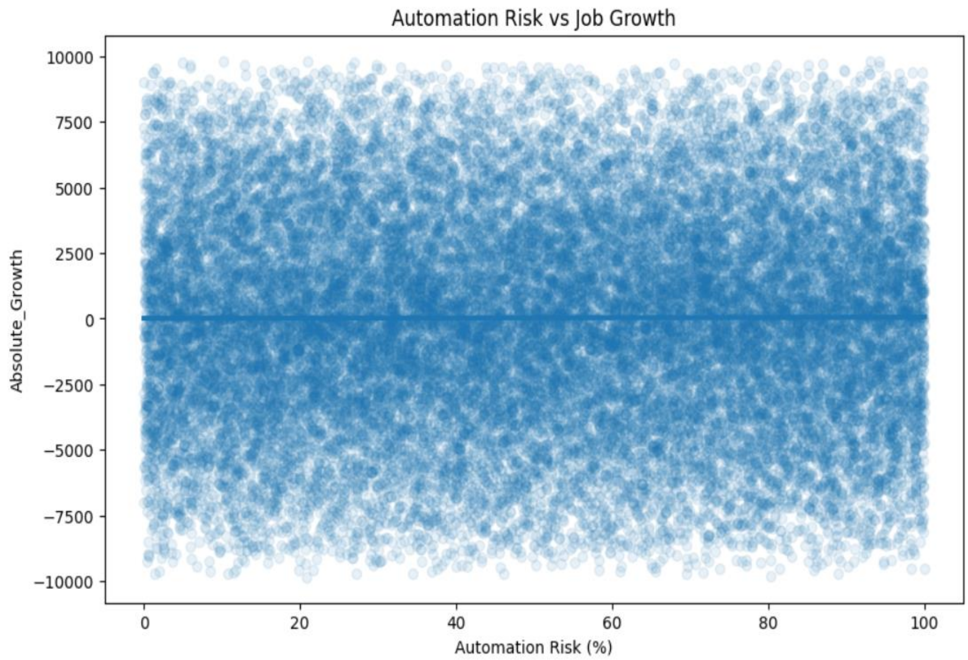


Figure 1. Relationship Between Automation Risk and Projected Employment Growth

Analysis of Automation Risk and Projected Employment Growth revealed virtually no relationship between the two variables (r = 0.003). Despite common assumptions that highly automatable jobs will experience declining demand, the dataset suggests that automation risk alone does not explain projected job growth. Additional factors such as industry, AI impact level, education requirements, and labor market demand will likely play a more significant role.

**Automation risk alone does not explain projected employment growth.**

**Finding 2: Do AI-Impacted Jobs grow Faster?**

```
Name: Absolute_Growth, dtype: float64
```

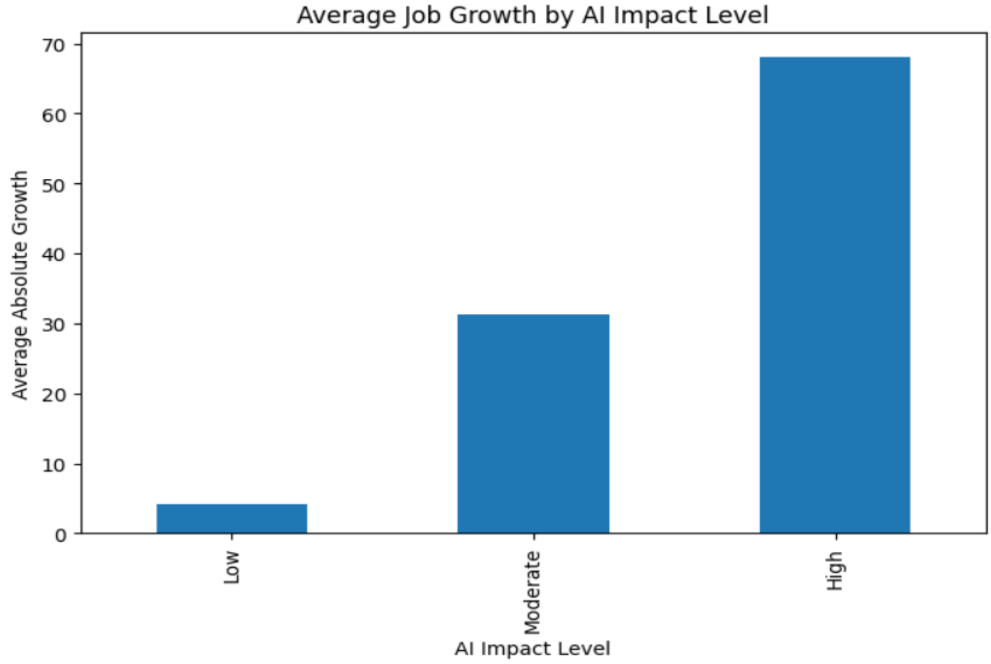


Figure 2. Average Projected Job Growth by AI Impact Level

Occupations classified as having High AI Impact exhibit the strongest projected employment growth, averaging approximately 68 additional openings compared with only 4 additional openings among Low AI Impact occupations. This finding suggests that AI may act as a workforce transformation driver rather than solely a job displacement mechanism, with AI-intensive occupations demonstrating the strongest projected demand through 2030.

**High AI Impact jobs tend to have higher projected growth. AI exposure appears associated with stronger projected workforce growth rather than workforce displacement.**

**Finding 3 – Industry-Specific Effects of AI**

| AI Impact Level | High | Low | Moderate |
|---|---|---|---|
| Industry | | | |
| **Education** | 74.616846 | -39.010260 | -108.288835 |
| **Entertainment** | 162.529366 | 151.890400 | -7.940789 |
| **Finance** | -94.114806 | 45.685524 | 82.113854 |
| **Healthcare** | 148.484634 | 77.890777 | 93.699052 |
| **IT** | 74.962097 | 90.907916 | 254.635910 |
| **Manufacturing** | 58.661994 | -36.397163 | -12.795699 |
| **Retail** | 81.331452 | -53.960599 | 158.258173 |
| **Transportation** | 36.621622 | -202.586743 | -198.239737 |

Figure 3.1. Industry-Specific Effects of AI on Projected Employment Growth.

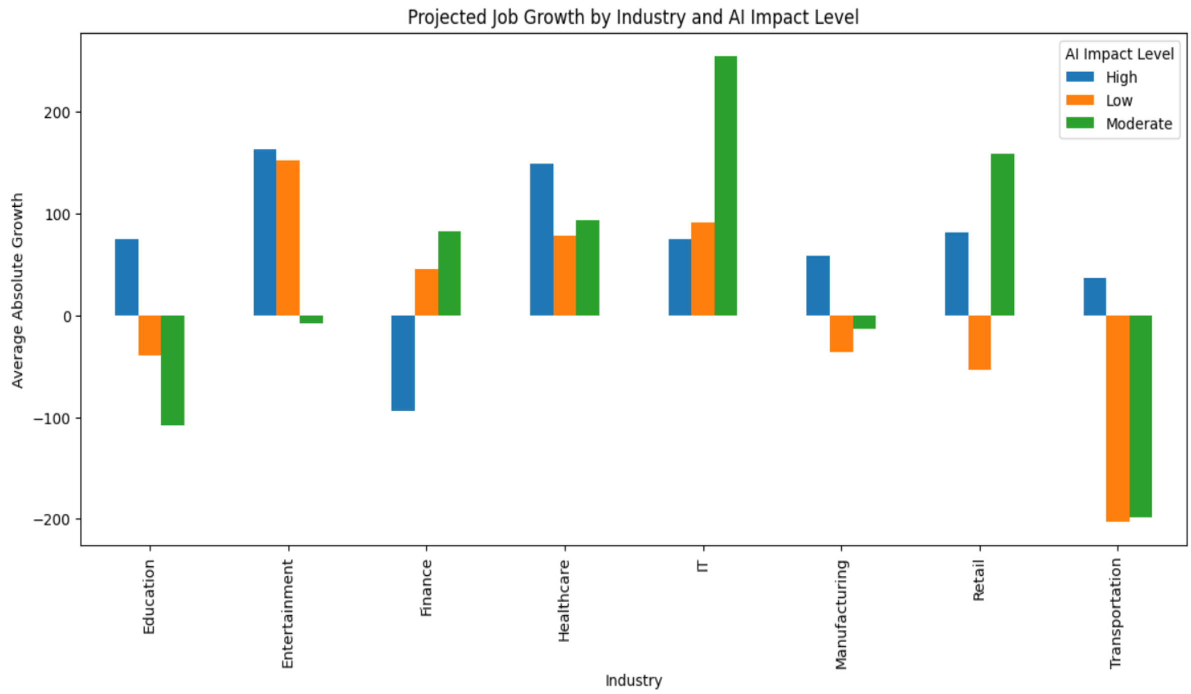


Figure 3. Industry-Specific Effects of AI on Projected Employment Growth

## Industry-Specific Effects of AI

Analysis of projected job growth by Industry and AI Impact Level revealed substantial variation across sectors.

Key findings include:

IT occupations with Moderate AI Impact exhibited the highest projected growth (+255 openings on average). Healthcare maintained growth across all AI Impact categories. Transportation showed the largest projected declines, particularly among Low and Moderate AI Impact occupations. Finance displayed mixed outcomes, suggesting uneven workforce transformation. Entertainment demonstrated strong growth despite anticipated AI disruption.

**These findings suggest that AI's workforce effects are industry-specific rather than universal, with some sectors experiencing expansion while others face contraction.**

**Industry characteristics appear more influential than automation risk alone.**

## Finding 4: Industry Growth Outlook (2024–2030)

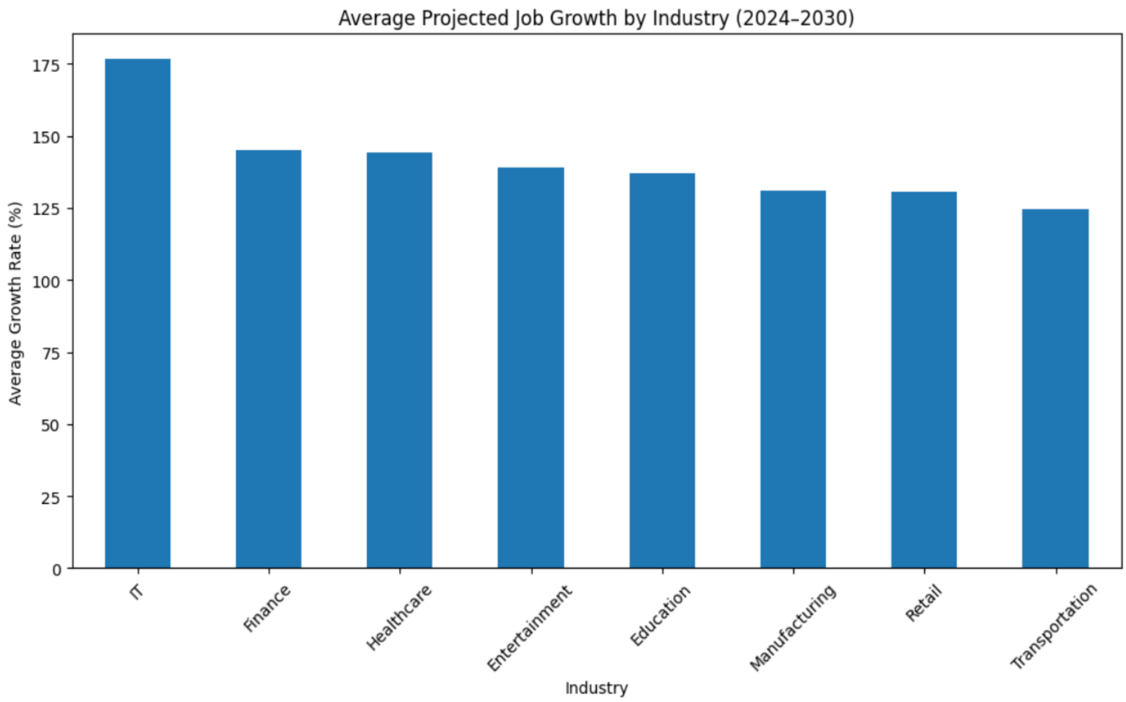


Figure 4. Projected Industry Growth Rates Between 2024 and 2030

## Industry Growth Analysis

The IT industry shows the highest projected growth rate (176.6%) between 2024 and 2030, significantly exceeding all other sectors.

Finance and Healthcare follow closely, both exceeding 144% projected growth.

Transportation shows the lowest projected growth (124.8%), though still positive.

A notable observation is that most industries fall within a relatively narrow growth range, suggesting that AI-driven workforce expansion may affect many sectors rather than being isolated to technology alone.

**The IT industry showing the strongest projected growth, suggests that AI is indeed creating demand for technical roles despite automation concerns.**

### Finding 5: Salary Analysis - Salary by AI Impact Level

**ANOVA Result**

F = 0.926

p = 0.396

ANOVA testing found no statistically significant difference in salaries.

```
AI Impact Level
High        90354.781024
Low         90265.040301
Moderate    89742.226538
Name: Median Salary (USD), dtype: float64
```

Figure 5. Average Salary by AI Impact Level

Despite substantial differences in projected job growth across AI Impact Levels, average salaries remained nearly identical. The largest difference between groups was less than 1%, suggesting that AI exposure is associated with employment demand rather than compensation.

## Limitations

Here is a list of some limitations I encountered in the analysis lifecycle:

- Dataset appears synthetic.
- Several variables exhibit near-zero correlations.
- Results should be interpreted as patterns within this dataset rather than real-world forecasts.

- Job Status labels were inconsistent with calculated growth metrics.

## Key Findings from the dataset

What does this finding mean to us:

1. Automation Risk does not predict employment growth (r = 0.003).
2. High AI Impact occupations experience substantially greater projected growth than Low AI Impact occupations.
3. AI effects vary significantly by industry as certain sectors experience strong workforce expansion.
4. AI-related work continues to expand - IT, Healthcare, and Entertainment appear best positioned for AI-driven expansion. IT shows the highest projected growth.
5. Some occupations face declining demand - Transportation appears most vulnerable to contraction but this is gradually expanding.
6. Salary levels remain consistent regardless of AI exposure.

## Strategic Recommendations

**Organizations**

- Prioritize workforce reskilling in Transportation and declining sectors.
- Invest in AI-enabled workforce development.

**Employees**

- Focus on AI-complementary skills.
- Consider growth sectors such as IT and Healthcare.

**Policymakers**

- Develop targeted transition programs for vulnerable industries

## CONCLUSION

The analysis suggests that AI is not simply eliminating jobs. Instead, it is redistributing demand toward roles and industries that can leverage AI effectively.